

SKEL 4223: Digital Signal Processing *with* MATLAB

CHAPTER 7

SPECTRUM REPRESENTATION OF DISCRETE-TIME SIGNAL

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1.0 Discrete-Time Fourier Transform (DTFT) Formulation

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1.1 Introduction

Formulation for discrete-time Fourier transform is given as:

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

where ω is called angular frequency. Instead of ω , DTFT can also be formulated based on normalized frequency f where $\omega = 2\pi f$ or $f = \omega/2\pi$.

Based on the formulation, DTFT will result a complex value. This is similar to z-transform since $z = e^{j\omega}$.

1.2 Single ω Value

When the DTFT is to be computed for a single value of ω , the following matrix operation can be used to solve the DTFT for signal $x[n]$.

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad \rightarrow \quad X = x \cdot E'$$

where $E = e^{-j\omega n}$ and x, n, E and X are all in the form of row vector. The sum operation is computed through the dot operation of the matrix multiplication.

For example, let's compute DTFT for signal $x[n] = [1 \ 2 \ 1]$ at $\omega = 0.1\pi$. From the signal, it is known that the $n = [0 \ 1 \ 2]$. Thus, the following is how the code looks like.

```
x = [1 2 1];           % Signal to be transformed
n = 0:length(x)-1;     % Discrete-time
w = 0.1*pi;            % Angular frequency
X = x*exp(-1i*w*n)';   % DTFT
```

X =

3.7111 + 1.2058i

1.3 Multiple ω values

When we need to compute DTFT for several values of ω at once, we need to set the ω as a vector. Then again, use matrix operation. Below is the matrix operation that is size compatible in MATLAB.

$$X = x.E', \quad \text{where } E = e^{-j\omega'n}$$

Here, E is a matrix while x , n , ω and X are all row vectors. Now, let's repeat the previous example with 2 values of ω as below. Since we are computing DTFT for 2 ω values, we get 2 complex values in the form of 2 elements row vector.

```
w = [0.1*pi 0.2*pi];
X = x*exp(-1i*w'*n)'
```

X =

3.7111 + 1.2058i 2.9271 + 2.1266i

This matrix operation is also works for single ω value. Thus, we can generally use the later matrix operation to any number of ω .

1.4 Plotting the DTFT

To plot the DTFT which is in frequency domain, we need to have enough number of ω samples or values where angular frequency ω will be the x-axis to the DTFT plot.

To demonstrate this, let's compute DTFT on 500 ω values within the range of $0 \leq \omega \leq \pi$. Note that previously it is shown that π is the maximum value of ω . In this example, `linspace()` function is used to generate linearly spaced vector of the ω .

For DTFT, the common practice is to plot the magnitude response, $|X(\omega)|$ and phase response, $\angle X(\omega)$. In this example we will only plot the magnitude response where:

$$|X(\omega)| = \left(X(\omega)_{real}^2 + X(\omega)_{imag}^2 \right)^{1/2}$$

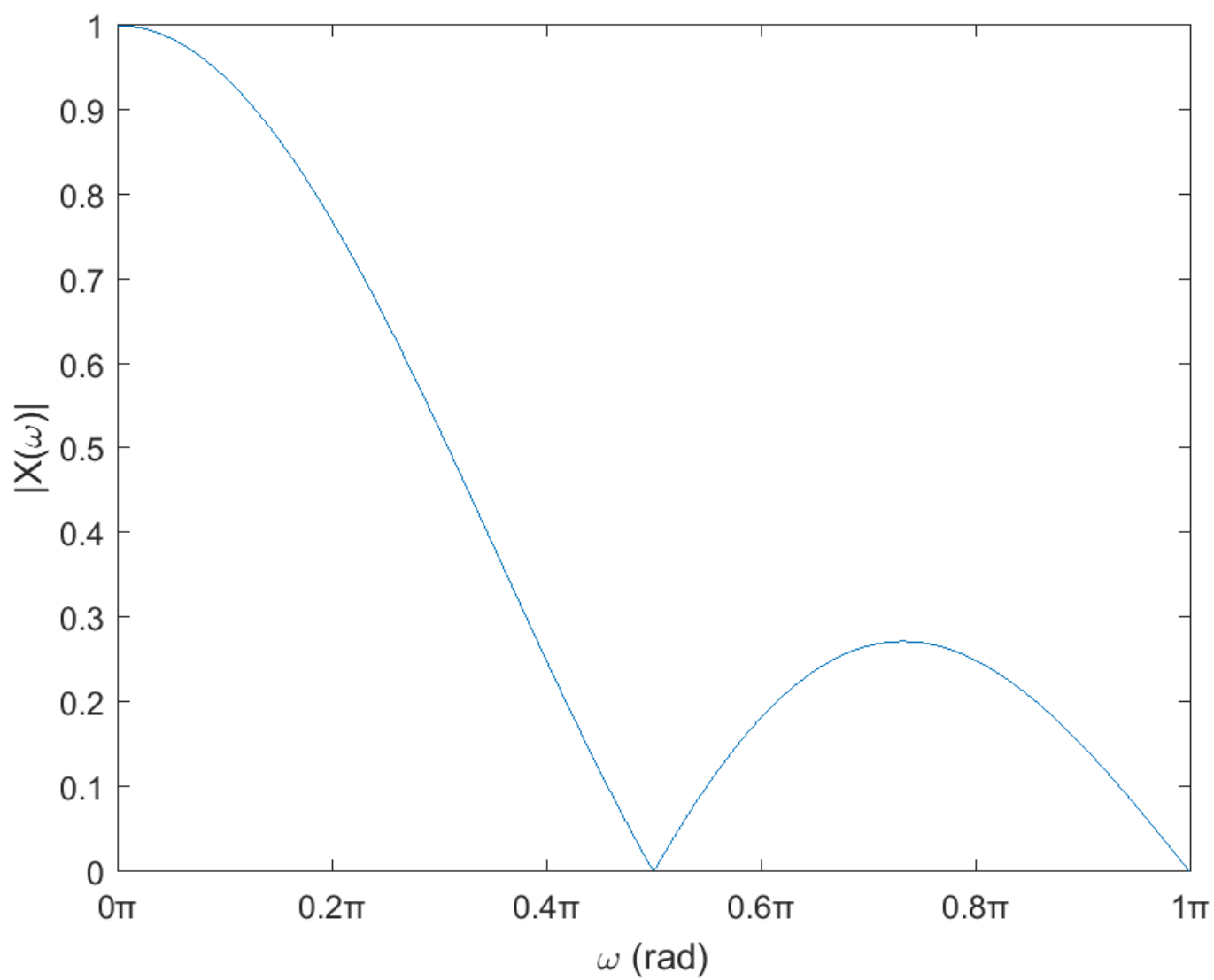
In MATLAB, magnitude of complex number can be computed using function `abs()`. The following is the code to plot the DTFT.

```

x = [0.25 0.25 0.25 0.25];           % Signal to be transformed
n = 0:length(x)-1;                   % Discrete-time
w = linspace(0,pi,500);              % Generate linearly spaced w

X = x*exp(-1i*w'*n)';                % DTFT
Xmag = abs(X);                       % Magnitude response

plot(w/pi,Xmag)
xtickformat('%g\texttimes 3C0')
xlabel('\omega (rad)')
ylabel('|X(\omega)|')
    
```



2.0 Frequency Conversion

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2.1 Introduction

In digital system, the only actual implemented Fourier Transform is Discrete Fourier Transform (DFT), denoted as $X[k]$ for discrete-time signal $x[n]$. This is because only DFT is computed with discrete frequency, k .

In MATLAB, instead of the formulation used in Section 1.0, DFT can be computed through Fast Fourier Transform (FFT) functions. FFT is a computationally optimised version of DFT. There are two functions computing the FFT for us to choose from, namely `fft()` and `freqz()`.

Other than DFT, the following type of Fourier Transform stay as either theoretical or implemented only on analog system:

1. Fourier Transform (FT): $X(F)$ or $X(\Omega)$.
2. Discrete-time Fourier Transform (DTFT): $X(f)$ or $X(\omega)$.

2.2 Type of Frequency

From the previous, we can see that there are five types of frequencies, listed as below:

1. Physical frequency, F in Hz
2. Angular frequency, Ω in $rad s^{-1}$
3. Normalised frequency, f with no unit
4. Angular frequency, ω in rad
5. Discrete frequency, k as sampled frequency.

Below is conversion formula from k to the other frequencies:

1. $F = F_s k / N$
2. $\Omega = 2\pi F_s k / N$
3. $f = k / N$
4. $\omega = 2\pi k / N$

Where F_s is the time-domain frequency sampling and N is the number of samples in frequency-domain. To have enough samples, N must be at least equals to the number of samples of the signal in time-domain.

2.3 Plotting FT and DTFT

Using the frequency conversion formulas, then we can plot FT and DTFT by mapping the k value on the x-axis to the FT and DTFT frequencies.

Now let's assume we have 1000 samples ($N = 1000$) in the frequency-domain. Then, $0 \leq k \leq N - 1$ and k is an integer number. Below is how we can convert the k into F , Ω , f and ω . Note that k for DFT is always start at $k = 0$. Also assume we have $F_s = 2 \text{ kHz}$.

```
Fs = 2000; N = 1000; k = 0:N-1;
```

```
x = [1 1 1];
```

```
X = fft(x,N);    % Do DFT, X[k]
```

```
% Frequency Conversion
```

```
F = Fs*k/N;
```

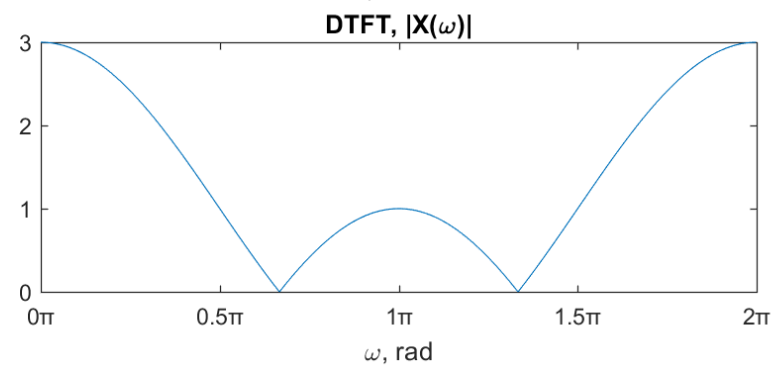
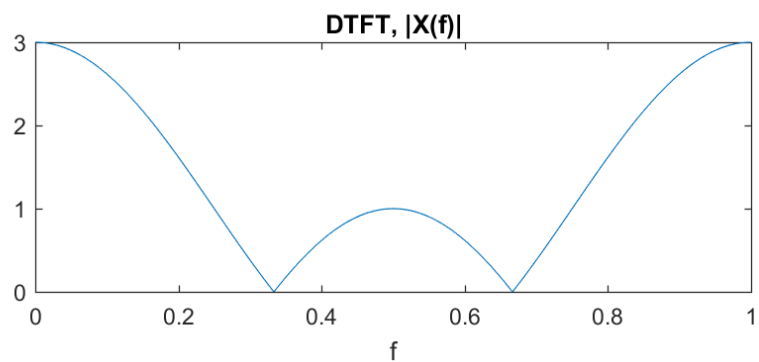
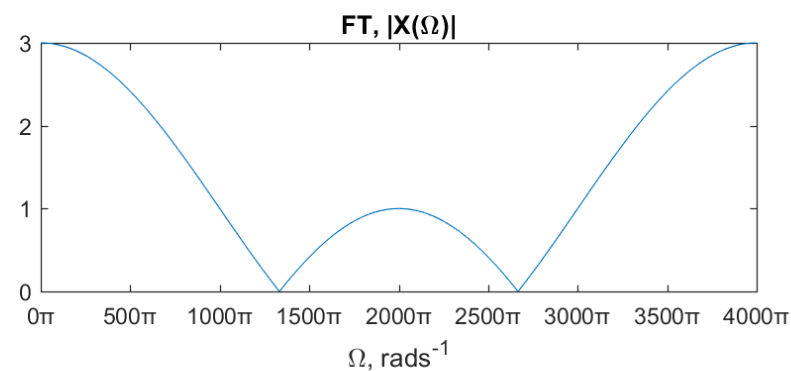
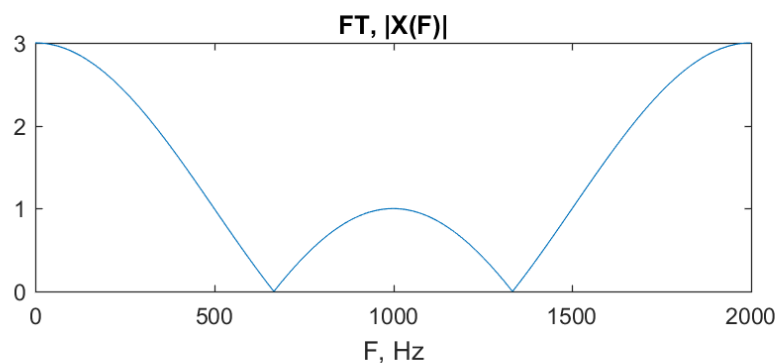
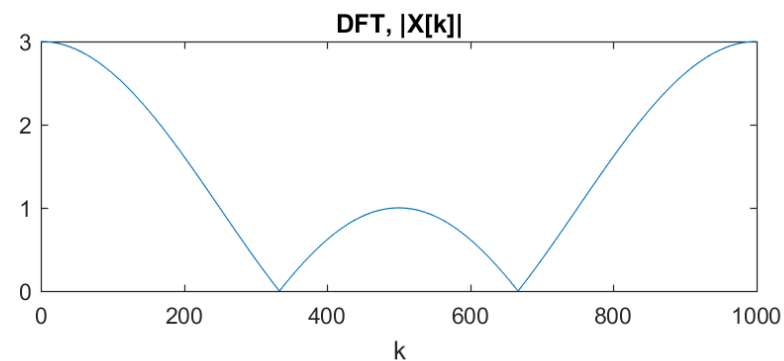
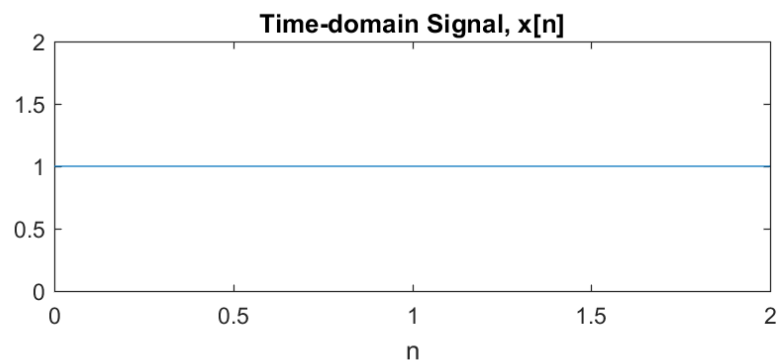
```
W = 2*pi*Fs*k/N;
```

```
f = k/N;
```

```
w = 2*pi*k/N;
```

Below is the code to plot the DFT, FT and DTFT. abs() function is used to plot the magnitude response of the spectrum. More discussion on the magnitude response will be covered in the next chapter.

```
subplot(3,2,1), plot(0:2,x)
title('Time-domain Signal, x[n]'), xlabel('n')
subplot(3,2,2), plot(k,abs(X))
title('DFT, |X[k]|'), xlabel('k')
subplot(3,2,3), plot(F,abs(X))
title('FT, |X(F)|'), xlabel('F, Hz')
subplot(3,2,4), plot(w/pi,abs(X))
title('FT, |X(\Omega)|')
xlabel('\Omega, rads^{-1}'), xtickformat('%g\x3C0')
subplot(3,2,5), plot(f,abs(X))
title('DTFT, |X(f)|'), xlabel('f')
subplot(3,2,6), plot(w/pi,abs(X))
title('DTFT, |X(\omega)|')
xlabel('\omega, rad'), xtickformat('%g\x3C0')
```

Note that plots for the FT are not accurate. We will discuss this in the next chapter.